Wei Ling Tan

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PROFILE

Year 2 Computer Science student focused on deep learning for medical images. I have trained and validated convolutional neural networks for retinal disease grading, with attention to calibration and clinician-facing explainability. Looking for a research project in healthcare AI.

EDUCATION

Nanyang Technological University, Singapore Aug 2024 - present

BSc Computer Science, College of Computing and Data Science. CGPA 3.99 / 5.00

Modules: SC4001 Neural Network and Deep Learning (A), SC1015 Introduction to Data Science and AI (A), SC3000 Artificial Intelligence (A-), SC4002 Natural Language Processing, CE4041 Pattern Recognition, SC3020 Database System Principles.

Self-study: Deep Learning Specialization (Coursera) and fast.ai Practical Deep Learning, with all projects completed.

RESEARCH AND PROJECT EXPERIENCE

Retinal Image Grading with CNNs Jan 2026 - Jun 2026

Self-initiated project, public datasets (APTOS 2019, EyePACS)

- Trained ResNet-50 and EfficientNet-B3 in PyTorch to grade diabetic retinopathy on the 5-level scale; reached 0.84 quadratic weighted kappa on a held-out set.
- Handled class imbalance with weighted sampling and focal loss, and poor-quality fundus images with CLAHE preprocessing and test-time augmentation.
- Calibrated the model with temperature scaling; expected calibration error fell from 0.11 to 0.03.
- Produced Grad-CAM and Score-CAM heatmaps and reviewed them with an ophthalmology student to check they highlight real lesions.

Research Attachment, CCDS (NTU) Jun 2025 - Aug 2025

Chest X-ray classification

- Trained DenseNet-121 on the school GPU cluster, tracked runs with Weights and Biases and wrote evaluation scripts (AUROC, per-class sensitivity).

HealthHack Singapore 2025 Mar 2025

2nd place

- Built a skin-lesion triage app with a MobileNetV3 model and an uncertainty flag that defers low-confidence cases to a doctor.

PUBLICATIONS AND PRESENTATIONS

- Poster: Calibrated CNN grading of diabetic retinopathy. NTU Undergraduate Research Showcase, 2026.

SKILLS

Languages: Python, SQL, C, Java. **ML:** PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, Grad-CAM, SHAP. **Tools:** Weights and Biases, Git, Linux, LaTeX.